Pediatric Bradycardia With Poor Perfusion

PC-02 | Pediatric Cardiac

Quick flow

01 Bradycardia + Poor Perfusion [EMT / AEMT / Paramedic]

HR <60 with hypotension, AMS, shock, weak pulses, or other poor perfusion.

02 Airway / Oxygenation First [EMT / AEMT / Paramedic]

Correct respiratory failure, obstruction, hypoxia, apnea/hypoventilation, hypoglycemia, hypothermia, acidosis, toxins, and other reversible causes.

03 HR Still <60 + Poor Perfusion? [EMT / AEMT / Paramedic]

After effective oxygenation and ventilation: begin CPR immediately, even with a pulse.

04 Vascular Access [AEMT / Paramedic]

Obtain rapid IV/IO without delaying CPR or effective ventilation; request Paramedic epinephrine for bradycardia with a pulse.

05 Paramedic - Epinephrine With Pulse [Paramedic]

Give 0.01 mg/kg IV/IO every 3-5 minutes while bradycardia with poor perfusion persists; maximum single dose 1 mg. Do not delay ventilation or CPR.

06 Vagal / AV-Block Cause? [Paramedic]

Atropine only for increased vagal tone or primary AV block; consider pacing for selected refractory conduction disease.

07 Reassess + Transport [EMT / AEMT / Paramedic]

Treat cause, monitor rhythm/perfusion, transition to arrest/ROSC pathway as needed, and notify pediatric-capable destination.

Overview

- For infants and children after the newly born period, pediatric bradycardia is usually caused by hypoxia; correct airway, oxygenation, ventilation, and reversible causes before medication.
- Begin CPR when heart rate remains below 60/min with poor perfusion despite effective oxygenation and ventilation.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Open and support the airway, oxygenate and ventilate effectively, assess glucose when authorized, and begin CPR for persistent HR below 60 with poor perfusion.
- Check pulse and response approximately every 2 minutes and transition immediately to the arrest pathway if pulseless.

AEMT

- Perform all EMT care plus IV/IO access, glucose analysis, crystalloid when hypovolemia is suspected, and authorized advanced airway support. Epinephrine for bradycardia with a pulse is Paramedic-only; transition to the AEMT/Paramedic pulseless-arrest epinephrine pathway only if the patient becomes pulseless.
- Do not delay CPR or ventilation for repeated IV attempts; use IO promptly when IV access is not rapidly available.

Paramedic

- Perform all prior care plus cardiac monitoring, weight-based epinephrine for bradycardia with a pulse, atropine for increased vagal tone or primary AV block, and transcutaneous pacing for selected refractory conduction disease.
- Consider immediate pacing for complete heart block or sinus-node dysfunction with persistent cardiopulmonary compromise when vascular access is not immediately available.
- If pacing is used in a conscious child, provide age-appropriate analgesia/sedation with Medical Control when time permits; do not delay life-saving pacing in extremis.

Indications

- Pediatric patient with bradycardia and hypotension, altered mental status, shock, poor perfusion, or impending arrest.

Treatment steps

- Treat the child, not the number: bradycardia is an emergency when associated with hypotension, altered mental status, shock, weak pulses, or other poor perfusion.
- If HR remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start high-quality CPR immediately even if a pulse is still present.
- Use continuous ECG, SpO_2, blood pressure, and waveform $EtCO_2$ after an advanced airway. Obtain a 12-lead if it does not delay resuscitation.
- Atropine is not a treatment for hypoxic bradycardia. Use it only for increased vagal tone or primary AV block.
- If the child becomes pulseless, transition immediately to PC-01/PC-04 Pediatric Pulseless Arrest. After sustained ROSC, move to PC-08 Pediatric Post-Resuscitation Care.

Medications

Epinephrine 0.1 mg/mL (1:10,000) - Paramedic only while pulse present

0.01 mg/kg IV/IO; maximum single dose 1 mg; repeat every 3-5 minutes while bradycardia with poor perfusion persists

Normal saline

10-20 mL/kg IV/IO only when hypovolemia is suspected; reassess perfusion, lungs, and cause after each bolus

Atropine

0.02 mg/kg IV/IO for increased vagal tone or primary AV block; minimum 0.1 mg; maximum single dose 0.5 mg child / 1 mg adolescent; may repeat once; maximum total 1 mg child / 2 mg adolescent

Warnings / precautions

- Do not delay oxygenation and ventilation for vascular access or medication.
- Atropine is not a treatment for hypoxic bradycardia.
- Atropine may be ineffective or harmful after cardiac transplantation and may cause paradoxical bradycardia.
- Interpret the rate in clinical context; asymptomatic bradycardia requires monitoring and treatment of the cause, not automatic medication.

Clinical pearls

- Bradycardia associated with poor perfusion may be a harbinger of pediatric cardiac arrest; ventilation and timely CPR are central interventions.
- Transcutaneous pacing is for selected refractory conduction disease, not routine hypoxic bradycardia.

Related pathways / contacts

AR-05 Pediatric Airway - Open for pediatric BVM and airway support.

PC-01 Pediatric Asystole / PEA - Open for nonshockable pulseless arrest.

PC-04 Pediatric Pulseless Arrest - Open immediately if the child becomes pulseless.

PC-08 Pediatric Post-Resuscitation Care - Open after sustained ROSC.

AO-02 Newly Born - Open for neonatal resuscitation immediately after birth.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

Revision: 2026-10-31 | Last verified: 2026-10-31